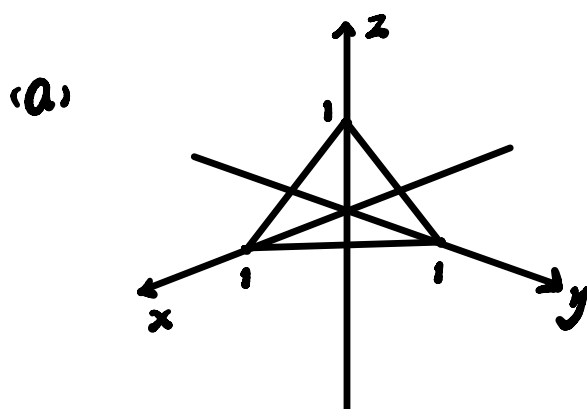


Progress test 3

1. Surface integral and integral theorem

- (a) Let $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ are vertices of a triangle S . If a vector field $\mathbf{F}$ is defined by $\mathbf{F}(x, y, z) = (1, e^z \cos x \sin y, -e^z \cos x \sin y)$, find $\iint_S \mathbf{F} \cdot d\mathbf{S}$. [30%]
- (b) Sketch a surface S given by $z = -x^2 - 4y^2 + 4$ for $z \geq 0$. If a vector field $\mathbf{G}(x, y, z) = xe^{xy}\mathbf{i} + 4ye^{xy}\mathbf{j} + z^3\mathbf{k}$, find $\iint_S (\nabla \times \mathbf{G}) \cdot d\mathbf{S}$. [30%]
- (c) Let a region B in $\mathbb{R}^3$ be a circular cylinder bounded by $(x - 5)^2 + (y - 2)^2 = 16$, $z = 4$, and $z = 8$. Sketch the region B . If a vector field $\mathbf{H}$ is given by $\mathbf{H}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, find $\iint_{\partial B} \mathbf{H} \cdot d\mathbf{S}$ where ∂B is the boundary surface of B . [40%]



$$x + y + z = 1$$

$$\therefore f = x + y + z = 1$$

$$\therefore \mathbf{n} = \frac{\nabla f}{\|\nabla f\|} = \frac{(1, 1, 1)}{\sqrt{1^2 + 1^2 + 1^2}} = \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$$

$$\therefore \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dA$$

$$= \iint_S (1, e^{\frac{\sqrt{3}}{3}} \cos(\frac{\sqrt{3}}{3}), -e^{\frac{\sqrt{3}}{3}} \cos(\frac{\sqrt{3}}{3})) \cdot \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right) dA$$

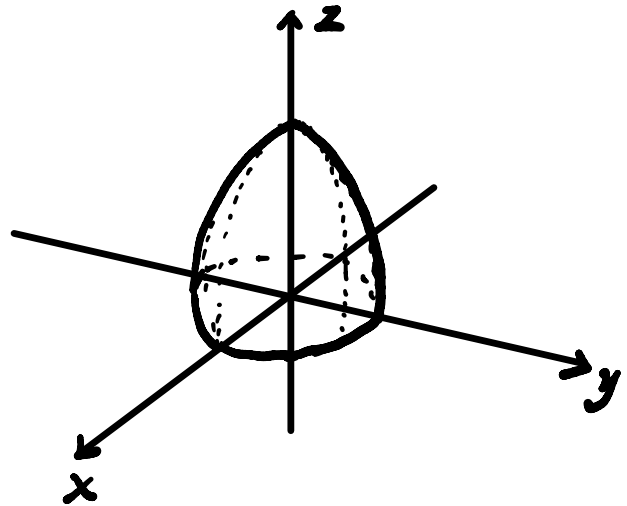
$$= \iint_S \frac{\sqrt{3}}{3} dA = \frac{\sqrt{3}}{3} A = \frac{\sqrt{3}}{3} \left(\frac{1}{2} \|a\| \|b\| \sin \theta\right) = \frac{\sqrt{3}}{6} \|a \times b\|$$

$$= \frac{\sqrt{3}}{6} \|(1, 0, 1) \times (0, 1, 1)\| = \frac{\sqrt{3}}{6} \|(1, -1, 1)\| = \frac{\sqrt{3}}{6} \cdot \sqrt{3} = \frac{1}{2}$$

(b) When $y = 0$, $z^2 = -x^2 + 4$

When $x = 0$, $z^2 = -4y^2 + 4$

When $z = 0$, $x^2 + 4y^2 = 4$



$$\therefore \iint_S (\nabla \times \mathbf{G}) \cdot d\mathbf{s}$$

$$= \int_{\partial S} \mathbf{G} \cdot d\mathbf{s}$$

At boundary condition, $z = 0 \therefore x^2 + 4y^2 = 4 = (r_1 \cos \theta)^2 + 4(r_2 \sin \theta)^2$

$$\therefore d\mathbf{s} = (2 \cos \theta, \sin \theta, 0)$$

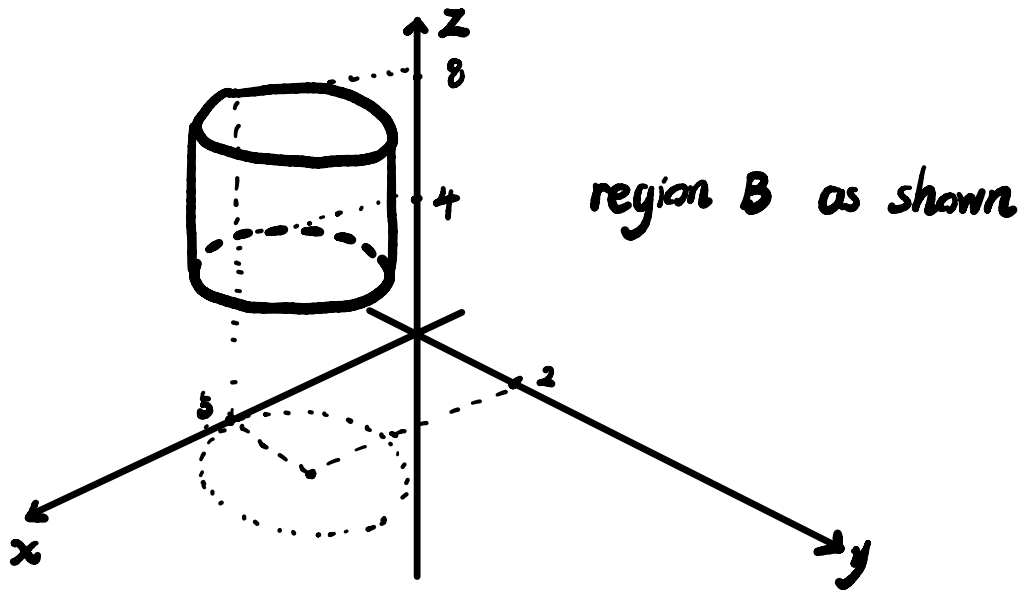
$$\therefore \iint_S (\nabla \times \mathbf{G}) \cdot d\mathbf{s} = \int_{\partial S} \mathbf{G} \cdot \frac{d(d\mathbf{s})}{d\theta} d\theta$$

$$= \int_0^{2\pi} (xe^{xy}, 4ye^{xy}, z^3) \cdot (-2 \sin \theta, \cos \theta, 0) d\theta$$

$$= \int_0^{2\pi} -4 \cos \theta \sin \theta e^{xy} + 4 \cos \theta \sin \theta e^{xy} + 0 d\theta$$

$$= \int_0^{2\pi} 0 d\theta = 0$$

(C)



$$\iint_{\partial B} \mathbf{H} \cdot d\mathbf{s} = \iiint_B \nabla \cdot \mathbf{H} \cdot dV$$

$$\text{for } z=4, \quad (x-5)^2 + (y-2)^2 = 16 = r^2 \quad \therefore r=4$$

$$\therefore \partial B = (4\cos\theta + 5, 4\sin\theta + 2, 4)$$

$$\text{for } z=8, \quad \partial B = (4\cos\theta + 5, 4\sin\theta + 2, 4)$$

$$\therefore \iiint_B \nabla \cdot \mathbf{H} \cdot dV = \iiint_B (1+1+1) dV = 3 \iiint_B dV$$

$$= 3 \cdot V = 3 \cdot (\pi r^2) \cdot h$$

$$= 192\pi$$